gGA Dimer Scans: U/t Dependence at $\Delta V = 0$ and $\Delta V = 2$

(Ghost Multiplicities $N_g = 2$ and $N_g = 4$, Direct-Minimization Solver)

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1 Total energy and error vs. exact: $N_g = 2$ and $N_g = 4$

Figures 1 and 2 show the variational energy E_{var}/t of the $N_g = 2$ and $N_g = 4$ solutions together with the exact (ED) energy, and the error $E_{\text{var}} - E_{\text{ED}}$ on a logarithmic scale, for $\Delta V = 0$ and $\Delta V = 2$ respectively. For $\Delta V = 2$ the plotted gGA energies are the cross-checked ("confirmed") solutions. The two N_g curves are indistinguishable on this scale. At $\Delta V = 0$, $\max |E(N_g{=}4) - E(N_g{=}2)| = 3 \times 10^{-8}$. At $\Delta V = 2$ the two agree to better than 10^{-6} for $U/t \lesssim 6.2$, while for $U/t \gtrsim 6.3$ the $N_g = 4$ energy is lower by up to 1.3×10^{-4} (at $U/t = 8.4$); this is still negligible against the error to the exact energy, ≈ 0.3–0.4 there (see Sec. 3).

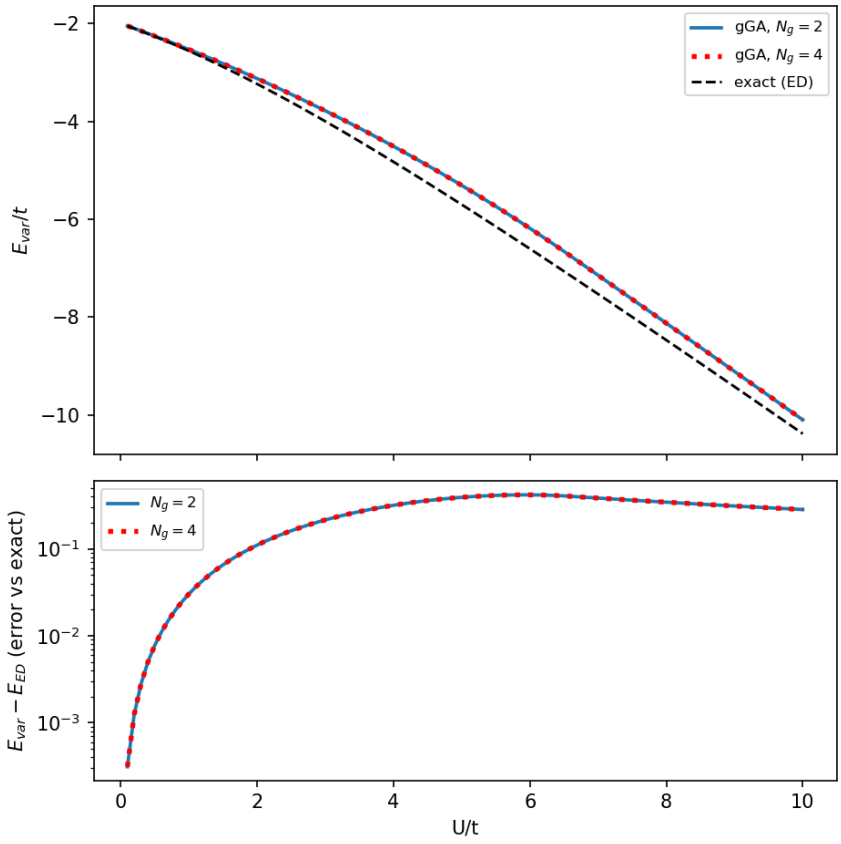


Figure 1: $\Delta V = 0$: E_{var}/t (top) and error $E_{\text{var}} - E_{\text{ED}}$ (bottom) vs. U/t for $N_g = 2$ and $N_g = 4$.

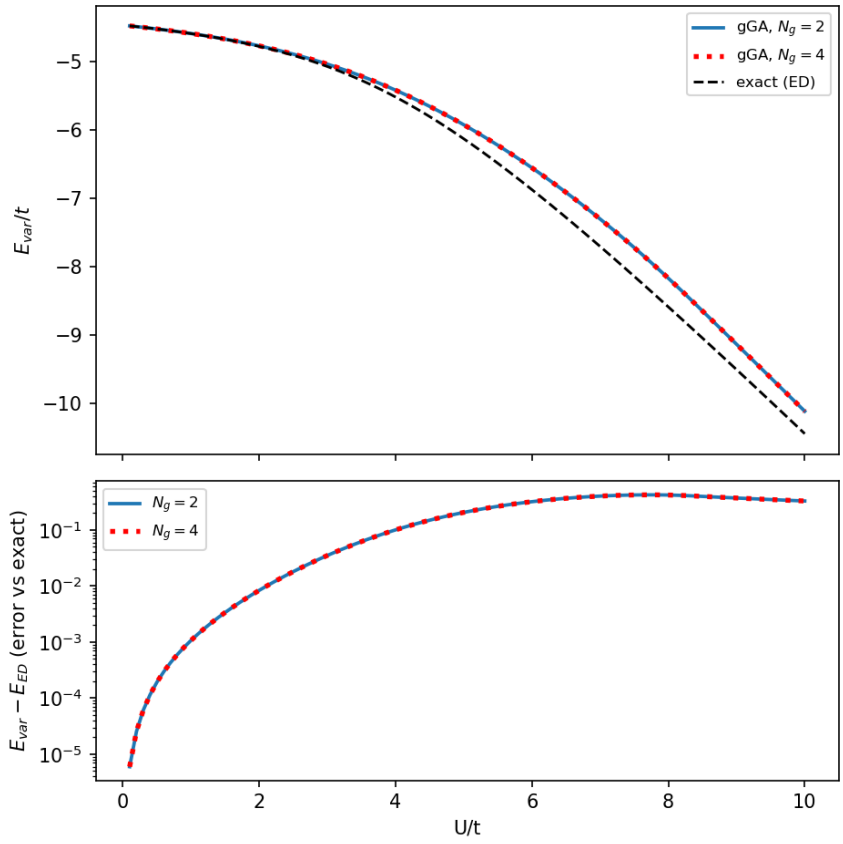
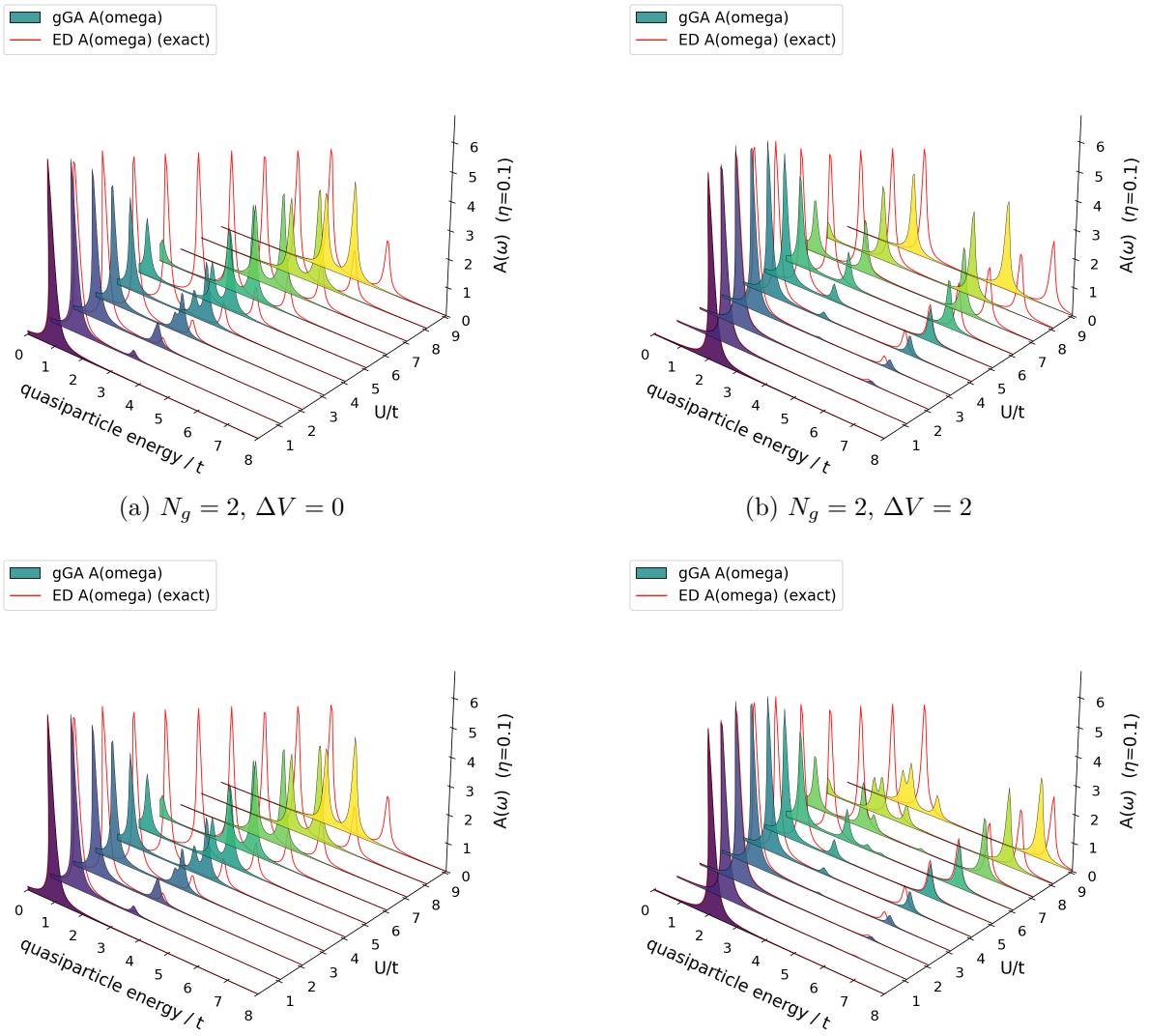


Figure 2: $\Delta V = 2$: E_{var}/t (top) and error $E_{\mathrm{var}} - E_{\mathrm{ED}}$ (bottom) vs. U/t for $N_g = 2$ and $N_g = 4$.

2 Spectra and spectral-function movies: $N_g = 2$ and $N_g = 4$

2.1 3D view

Figure 3 shows, for each $(N_g, \Delta V)$ combination, the gGA spectral function $A(\omega)$ (filled ridgelines, built from the quasiparticle poles and weights of the converged solution, broadening $\eta = 0.1$) as a function of U/t, together with the exact (ED) spectral function (red lines). Rows are $N_g = 2$ and $N_g = 4$, columns are $\Delta V = 0$ and $\Delta V = 2$. The $\Delta V = 2$ panels use the cross-checked solutions.



(a) $N_g = 2$, $\Delta V = 0$

(b) $N_g = 2$, $\Delta V = 2$

(c) $N_g = 4$, $\Delta V = 0$

(d) $N_g = 4$, $\Delta V = 2$

Figure 3: gGA spectral function $A(\omega)$ (filled) vs. ED (red) as a function of U/t for $N_g = 2$ (top) and $N_g = 4$ (bottom), at $\Delta V = 0$ (left) and $\Delta V = 2$ (right). Only $\omega \geq 0$ is shown: $A(\omega) = A(-\omega)$ in both gGA and ED.

Analysis. The number of quasiparticle branches is $2(1 + N_g)$: six for $N_g = 2$ and ten for $N_g = 4$ (not a doubling). In all four panels the gGA peaks coincide with the ED ones at small U/t and move away from them as U/t grows, consistent with the error in the total energy growing with U/t (Sec. 1). The $N_g = 4$ ridgelines are visually indistinguishable from the $N_g = 2$ ones at the same ΔV: the extra branches carry no visible spectral weight, in line with the energies, so the gGA peaks do not follow the ED peaks more closely over the scanned range; the additional branches are the ones discussed as "sleeping" ghosts in Sec. 3. The effect of ΔV is visible in the same way for both N_g: at $\Delta V = 2$ the branches are well separated at small U/t and bunch together as U/t increases, with a rearrangement between $U/t = 0.6$ and 0.7 (Sec. 3).

2.2 Spectral function movies

- $N_g = 2$, $\Delta V = 0$: spectral-function movie
- $N_g = 2$, $\Delta V = 2$: spectral-function movie
- $N_g = 4$, $\Delta V = 0$: spectral-function movie
- $N_g = 4$, $\Delta V = 2$: spectral-function movie

(Links open the movie with the PDF viewer's associated application, if the viewer supports `run:` links.)

Quasiparticle-weight movies. The same scans are shown level by level: for each U/t the eigenvalues ϵ_n^{qp} of H_{qp} with the weight carried by each level (summed over both sites and spins, so the total weight is that of $A(\omega)$), and the ED poles as red open circles (linear and logarithmic scales; script `make_qp_weights_video.py`). A level with weight $\geq 10^{-3}$ is a peak (levels without weight are light gray). At $N_g = 2$ a peak keeps its color along U/t (consecutive U/t points are matched by closeness in energy and weight), and the colors **blue**, **orange**, **green** follow the order of increasing energy at the U/t where all the peaks coexist. At $N_g = 4$ a peak that corresponds to an $N_g = 2$ peak (matched at the same U/t) has its color, so the same physical peak has the same color at $N_g = 2$ and 4; the extra $N_g = 4$ peaks are **purple** and **brown**, in order of increasing energy.

- $N_g = 2$, $\Delta V = 0$: quasiparticle-weight movie
- $N_g = 2$, $\Delta V = 2$: quasiparticle-weight movie
- $N_g = 4$, $\Delta V = 0$: quasiparticle-weight movie
- $N_g = 4$, $\Delta V = 2$: quasiparticle-weight movie

Weight transfer. Positive-energy side. Each peak is written as (weight, energy) in the color of its peak in the movies (weights summed over sites and spins, total 2.0; ED poles in red). At $\Delta V = 0$ the peaks are the blue one, which carries almost all the weight at small U/t, and the orange and green ones, which are born at $U/t \simeq 0.5$ near $\epsilon = 3$. As U/t grows the blue peak loses weight to the orange and green ones while its energy goes to zero, and both vanish at a critical U_c/t ($N_g = 2$ and 4 agree; ED has no such event; the values come from a linear extrapolation of the last grid points); the orange and green peaks continue, with the same colors, above U_c.

- $\Delta V = 0$, $U/t \lesssim 1$ ($U/t = 1$): (1.92, 0.97) carries $\geq 96\%$; (0.02, 2.99) and (0.06, 3.02) hold the rest. ED: (1.97, 1.06), (0.03, 3.06).
- $\Delta V = 0$, $1 \lesssim U/t < U_c/t \simeq 6.56$: the blue peak loses its weight and the other two gain it. $U/t = 4$: (1.08, 0.57), (0.27, 2.83), (0.66, 3.26); $U/t = 6$: (0.27, 0.14), (0.66, 2.56), (1.05, 3.41); $U/t = 6.5$: (0.03, 0.02), (0.80, 2.46), (1.13, 3.43). ED at $U/t = 4$: (1.71, 1.83), (0.29, 3.83).
- $\Delta V = 0$, $U/t > U_c/t$: only the orange and green peaks. $U/t = 7$: (0.83, 2.69), (1.13, 3.67); $U/t = 10$: (0.89, 4.29), (1.10, 5.29). ED at $U/t = 10$: (1.37, 4.39), (0.63, 6.39).

At $\Delta V = 2$ the blue peak is again the main one; the orange peak is the middle one (born at $U/t \simeq 1.5$–3) and the green one the upper one (born at $U/t \simeq 1$).

- $\Delta V = 2$, $U/t \lesssim 3$ ($U/t = 3$): (1.89, 1.25), (0.015, 3.59), (0.10, 5.99); ED: (1.89, 1.34), (0.11, 5.81).
- $\Delta V = 2, 3 \lesssim U/t < U_c/t \simeq 8.3$: the blue peak loses weight and the other two gain it. $U/t = 5$: (1.46, 0.81), (0.18, 2.57), (0.36, 5.90); $U/t = 6$: (1.10, 0.59), (0.37, 2.33), (0.53, 5.83); $U/t = 8$: (0.18, 0.09), (0.84, 2.08), (0.95, 5.28). The blue peak has weight 0.02 at $U/t = 8.3$ and 10^{-3} at 8.5 (it stays at $\epsilon \lesssim 10^{-3}$).

- $\Delta V = 2$, $U/t > U_c/t$ ($N_g = 2$): only the orange and green peaks. $U/t = 9$: (0.94, 2.38), (1.02, 5.66); $U/t = 10$: (0.96, 2.88), (1.02, 6.36). ED at $U/t = 10$: (1.22, 3.21), (0.78, 7.68).

Above U_c the two weights approach each other but are not equal up to $U/t = 10$; gGA is closer to equal sharing than ED, i.e. it puts too much weight on the upper level. For $N_g = 4$ there are four more levels, near $\epsilon = 0$ and without weight at small U/t (colors as in the movies). At $\Delta V = 0$ the peaks are those of $N_g = 2$ (same colors and values, differences $\leq 2 \times 10^{-3}$) for all U/t, including U_c/t: the extra levels never carry weight. At $\Delta V = 2$ the regions are:

- $\Delta V = 2$, $U/t \lesssim 5.3$: as $N_g = 2$ (the smallest peaks differ by $\lesssim 6 \times 10^{-3}$ in weight).
- $\Delta V = 2$, $5.4 \lesssim U/t < 7.5$: two extra peaks appear, purple and brown, and come and go with U/t. $U/t = 6$: (1.09, 0.59), (0.34, 2.27), (0.05, 3.01), (0.03, 3.26), (0.50, 6.15), against (1.10, 0.59), (0.37, 2.33), (0.53, 5.83) at $N_g = 2$; $U/t = 7$ (brown below the weight threshold): (0.66, 0.35), (0.53, 2.15), (0.16, 2.56), (0.66, 6.31), against (0.66, 0.35), (0.60, 2.18), (0.73, 5.65). The purple peak takes its weight from the orange and green ones, and the blue one is not changed.
- $\Delta V = 2$, $U/t = 7.5 \to 7.6$: the solution changes discontinuously (five peaks to four, the energy of the green peak jumps from 6.38 to 5.48).
- $\Delta V = 2$, $7.6 \lesssim U/t < U_c/t \simeq 8.3$: five peaks; $U/t = 8$: (0.17, 0.09), (0.38, 1.79), (0.43, 2.08), (0.18, 3.06), (0.85, 6.44). The blue peak vanishes at about U_c (0.014 at $U/t = 8.3$), as at $N_g = 2$.
- $\Delta V = 2$, $U/t > U_c/t$: four peaks. $U/t = 9$: (0.35, 2.00), (0.46, 2.29), (0.27, 3.22), (0.92, 6.82); $U/t = 10$: (0.34, 2.53), (0.58, 2.85), (0.15, 4.14), (0.93, 7.32), against (0.96, 2.88), (1.02, 6.36) at $N_g = 2$ and (1.22, 3.21), (0.78, 7.68) in ED. The weight of the orange peak of $N_g = 2$ (0.96) is shared by the orange peak and the two extra ones (1.07 in total), and the green peak moves toward ED (7.32 against 6.36 at $N_g = 2$; ED 7.68) and loses weight (0.93, ED 0.78).

The $N_g = 4$ departure from $N_g = 2$ starts at $U/t \simeq 5.4$, consistent with the $N_g = 4$ energy departing from the $N_g = 2$ one for $U/t \gtrsim 6.2$ (Sec. 1).

3 Discussion: $\Delta V = 0$ vs. $\Delta V = 2$

Weak coupling: ΔV dominates. At small U/t the two scans differ substantially. At $U/t = 0.1$, $E_{\text{var}}/t = -2.0503$ for $\Delta V = 0$ but -4.4823 for $\Delta V = 2$ (both N_g). This is the expected $U \to 0$ limit: with $\Delta V = 0$ the dimer's noninteracting energy is $-2t = -2$ plus a small $O(U)$ correction, while at $\Delta V = 2$ the site-splitting opens a one-body level gap and the $U \to 0$ energy approaches the closed form $E = -2\sqrt{t^2 + \Delta V^2} = -2\sqrt{5} \approx -4.472$. The gap between the two curves is largest here, $\Delta E \approx 2.43$, because ΔV acts as an unscreened one-body potential that the (still weak) correlations barely renormalize.

Strong coupling: the curves converge. As U/t grows the two energies draw close together: at $U/t = 10$, $E_{\text{var}}/t = -10.101$ ($\Delta V = 0$) vs. -10.120 ($\Delta V = 2$), a difference of only ≈ 0.02 – two orders of magnitude smaller than at weak coupling. In the Mott-like regime the total energy is governed almost entirely by U (both sites approach single occupancy), and the fixed $\Delta V = 2$ site-splitting becomes a small perturbation on top of the dominant Hubbard physics. The quasiparticle spectra (3D plots, Fig. 3) show the same trend: the $\Delta V = 2$ branches, well separated at small U, bunch together with the $\Delta V = 0$-like structure as U/t increases.

Level crossing at $\Delta V = 2$. The $\Delta V = 2$ quasiparticle energies show a sharp rearrangement between $U/t = 0.6$ and $U/t = 0.7$ (e.g. $N_g = 2$: $\epsilon^{qp} \approx -2.00 \to -3.12$), visible as a kink in the

3D plots; this is the level crossing where the ΔV-induced one-body gap stops being the dominant scale and the Hubbard repulsion takes over.

Ghost multiplicity ($N_g = 2$ vs. $N_g = 4$). At both $\Delta V = 0$ and $\Delta V = 2$, doubling the ghost multiplicity from $N_g = 2$ to $N_g = 4$ leaves the lowest variational energy numerically unchanged at $\Delta V = 0$ over the whole scanned range, and at $\Delta V = 2$ for $U/t \lesssim 6.2$ (for larger U/t the $N_g = 4$ energy is lower by at most 1.3×10^{-4}, Sec. 1): at $\Delta V = 0$, $E(N_g{=}4) - E(N_g{=}2) = 0$ to the printed precision at essentially every U/t (largest deviation $\sim 3 \times 10^{-8}$, at the level of solver tolerance), and the confirmed energies show the same agreement at $\Delta V = 2$ (e.g. $U/t = 1.0$: -4.593786 vs. -4.593786). The extra ghost degrees of freedom introduced at $N_g = 4$ therefore stay essentially energetically inert for this dimer at both values of ΔV in this U range – consistent with the known $N_g \geq 2$ "sleeping ghost" caveat: the additional branches exist in the quasiparticle spectrum (compare Fig. 3, which shows ten instead of six ϵ_n^{qp} branches at $N_g = 4$) but do not lower the ground-state energy below what $N_g = 2$ already finds.